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Combining Unsupervised and Supervised Deep Learning for Alzheimer's Disease Detection by Fractional Anisotropy Imaging

Giovanna Castellano
Dept. of Computer Science
University of Bari
Bari, Italy
giovanna.castellano@uniba.it

Eufemia Lella
Innovation Lab
Exprivia S.p.A.
Molfetta, Italy
eufemia.lella@exprivia.com

Valerio Longo
Dept. of Computer Science
University of Bari
Bari, Italy
v.longo20@studenti.uniba.it

Giuseppe Placidi
A^2VI-Lab, c/o MeSVA Department
University of L'Aquila
L'Aquila, Italy
giuseppe.placidi@univaq.it

Matteo Polsinelli
Dept. of Computer Science
University of Salerno
Fisciano, Italy
mpolsinelli@unisa.it

Gennaro Vessio
Dept. of Computer Science
University of Bari
Bari, Italy
gennaro.vessio@uniba.it

***Abstract*—We propose a new approach for Alzheimer's disease (AD) detection using diffusion tensor imaging, specifically fractional anisotropy (FA) images, based on a combination of unsupervised and supervised deep learning techniques. Our method involves training a 3D convolutional autoencoder to learn low-dimensional representations of FA images in an unsupervised manner and using the learned representations to pre-train a supervised 3D convolutional classifier to predict the presence or absence of AD. Unsupervised pre-training can improve the classifier's performance, especially when difficult-to-collect labeled data are limited. We evaluate our approach on the OASIS-3 dataset and demonstrate promising performance.**

***Index Terms*—3D convolutional neural networks, Alzheimer's disease, autoencoder, deep learning, DTI, fractional anisotropy**

I. Introduction

Alzheimer's disease (AD) is a progressive and debilitating neurological disorder affecting millions of people worldwide, the early detection of which is critical for timely treatment and improved patient outcomes [1]. In this context, diffusion tensor imaging (DTI) is an advanced magnetic resonance imaging (MRI) technique that has proven helpful for detecting early signs of AD by measuring the integrity of white matter fiber tracts in the brain [2]. In particular, fractional anisotropy (FA) is a widely used metric derived from DTI, reflecting the degree of anisotropy in water diffusion, which has been found to be altered in AD patients in the literature, e.g. [3].

However, the analysis of FA images can be challenging due to the complexity and variability of these data. Traditional image analysis methods have limitations in capturing subtle patterns in the images and require extensive feature engineering and manual intervention, which can be time-consuming and error-prone. On the other hand, recent advances in deep learning techniques have shown great promise in various medical imaging applications, including AD detection by DTI, e.g. [4], [5]. Deep learning models can learn to automatically extract meaningful features from raw neuroimaging data and have the potential to improve the accuracy and efficiency of AD diagnosis [6].

This paper proposes a novel approach combining unsupervised and supervised deep learning techniques for AD detection using FA images. Our method involves training a 3D convolutional autoencoder to learn low-dimensional representations of FA images in an unsupervised manner. The learned representations are then used as a pre-training method of a 3D convolutional neural network (CNN) to predict the presence or absence of AD. 3D CNNs are powerful deep learning models that, unlike traditional 2D CNNs, can learn spatial features directly from volumetric data. The use of an autoencoder for unsupervised pre-training can help improve the performance of the subsequent supervised learning task and alleviate the need for a large annotated dataset with difficult-to-collect labels.

We evaluated our approach on the OASIS-3 dataset, a recently released and publicly available dataset for AD [7], obtaining promising performance highlighting the effectiveness of combining unsupervised and supervised deep learning to address this task. The proposed method has the potential to provide a quantitative and automated method for detecting AD, along with other diagnostic tools, in clinical practice.

The rest of the paper is structured as follows. Section II presents an overview of related work. Section III describes the material used in this study. Section IV discusses the proposed method. Section V presents the experimental results. Finally, Section VI concludes the paper.

II. Related Work

Alzheimer's disease detection has been an active area of research in medical imaging for many years. In particular,

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recent advances in medical imaging technology have allowed researchers to investigate the use of deep learning methods for this task. Deep learning approaches offer the potential for new solutions for AD detection, which could significantly improve the efficiency and accuracy of clinical practice over standard machine learning techniques.

Most of the work on AD detection has focused on MRI [8], [9]. MRI is a noninvasive and widely used imaging modality that can provide detailed structural information about the brain. For this reason, it has been widely studied for its potential to detect structural changes due to AD. In recent years, deep learning techniques have been applied to MRI data for automated detection and diagnosis of AD and CNNs have shown promising results in this regard. One approach to using CNNs for AD detection is to fit pre-trained models that were initially trained on large natural image datasets. These pre-trained models have already learned low-level features, such as edges and textures, that can be useful for the subsequent task. In fact, using *transfer learning*, researchers can refine these pre-trained models on MRI data for AD detection [10]–[12]. Another approach is to design custom CNN architectures specifically for AD, e.g. [13]. In addition to 2D CNNs, 3D CNNs have also been studied for AD detection using MRI data. 3D CNNs can capture the spatial relationships between voxels in 3D MRI volumes, which may be essential for the task at hand. Some studies have used 3D CNNs pre-trained on natural images, while others have used a pre-training based on autoencoders [14]–[16].

Concerning DTI [17], it is an imaging technique that measures the diffusion of water molecules in biological tissues in multiple directions to estimate the directionality and extent of water diffusion, enabling the reconstruction of white matter tracts in the brain. To this end, it has been widely used to study brain connectivity and white matter integrity and has also been applied to diagnosing neurodegenerative diseases, including AD. Traditional machine learning techniques combined with tractography-based approaches have been applied to DTI data for AD characterisation [18] and diagnosis at different stages of cognitive decline [19]. Other approaches consist of feeding traditional machine learning algorithms with features extracted by manually selecting regions of interest from DTI data [20], [21] or by selecting voxels through spatial-based statistics [22]. A further approach is to use principal component analysis to reduce the dimensionality of DTI data and extract features for the same objective [23].

Recent studies have shown that combining information from multiple imaging modalities, such as structural MRI and diffusion imaging, can improve the accuracy of AD diagnosis. These multimodal approaches can provide complementary information on brain structure and function that may be useful in identifying early biomarkers of AD. Wang et al. [24] used a 2D CNN on both the structural and functional connectivity networks extracted from DTI and functional MRI, respectively. They analyzed the performance of each connectivity network alone and in combination and found that combining both networks resulted in better accuracy for AD detection. In addition, other studies have applied deep learning on multimodal imaging data focusing on the hippocampal region, which is known to be affected in the early stages of AD [25], [26].

While many studies have been on using deep learning with structural MRI data for AD diagnosis, fewer studies focused on DTI data in a deep learning setting. Marzban et al. [27] attempted to solve the problem of AD classification using a small 2D CNN with three different maps from DTI: fractional anisotropy, mean diffusivity, and mode of anisotropy, along with the gray matter map from structural MRI. They accurately classified AD patients and healthy controls using these features. Similarly, Houria et al. [28] used a custom 2D CNN architecture with an SVM-based classification head to select slices from two different DTI maps (fractional anisotropy and mean diffusivity) and the gray matter map from structural MRI. They extracted features using these three models and combined them using an SVM to produce the final output.

The novelty of our approach lies in the combination of unsupervised and supervised learning methods using 3D CNNs to detect AD with FA images. This could significantly contribute to improved performance. In addition, the use of 3D CNNs is relatively new, as many studies have focused only on 2D CNNs. Finally, we used the recently published OASIS-3 dataset, one of the largest and most comprehensive publicly available datasets for AD detection.

III. Materials

This study was primarily based on the longitudinal OASIS-3 dataset [7], which consists of a collection of MRI and PET scans for 1098 subjects, of whom 609 are cognitively normal, and 489 are in different stages of cognitive decline. For each subject, one or more sessions have been performed over time, and some sessions have more than one scan available. For this work, we considered the 1754 DWI sessions, corresponding to 3255 scans. In addition, to further increase the number of samples, we used the IXI dataset [29], which provides 400 DWI sessions of cognitively normal subjects. While DWI refers to the contrast of acquired images, DTI is a specific type of DWI modeling [30]. It allows the measurement of other diffusion parameters, such as fractional anisotropy.

Each available DWI was analyzed to find corrupted scans and sessions with a mismatch between the *b*-values, the *b*-vectors, and the corresponding number of volumes. The *b*-values and *b*-vectors describe each volume's magnetic field strength and direction. We also discarded DWIs with less than ten volumes (shells). After this cleaning step, the available DWI scans became 3124. To provide a ground truth, we used the clinicians' judgments along with the clinical diagnoses provided with the dataset to infer a diagnosis for each subject. From the clinicians' judgments, we obtained the age from which the subject first manifested signs of mental illness. At the same time, the clinical diagnoses provide information about the subject's illness, taking into account the subject's history. Each subject who has undergone one or more visits has both the judgment and the diagnosis corresponding to the visit; because of the degenerative nature of AD, we considered

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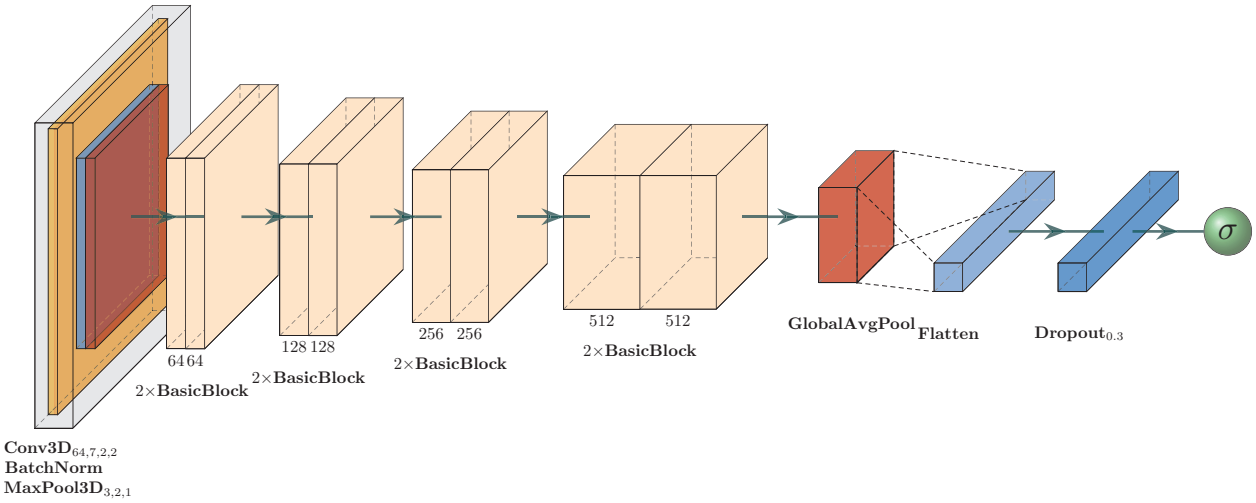

Fig. 1. Schema of the classification model. The number under each basic block represents the number of filters in that block. For Conv3D and MaxPool, the subscripts represent the number of filters, kernel size, stride, and padding, respectively.

the last available diagnosis and assigned to the subject a label of 0 (negative) in case of a healthy subject or non-AD related disease and 1 (positive) in case of AD. This labeling strategy allowed us to set up the problem as a binary classification task. Unfortunately, not all subjects have a visit entry: these subjects (and the corresponding scans) cannot be used for classification and will be part of the dataset used for autoencoder training. Ultimately, we obtained 2638 different scans (1531 unique sessions) belonging to 863 subjects, of whom 173 were AD.

The dataset is highly unbalanced (about 98% negative samples). To mitigate this concern, we subsampled on the negative scans. To do this, we used a strategy based on the k-means algorithm, in which the number of clusters k equals the number of samples in the minority class, and the majority class samples are then grouped into k clusters. A sample (i.e., the medoid of the cluster) is taken from each cluster. Since the goal was to make our model robust in detecting AD, we kept as minority class samples all scans belonging to both AD and unhealthy subjects, then we subsampled only the healthy ones; the features used for clustering were the age at which the scan was taken and the gender of the subject. After this step, the dataset contains 491 negative samples (related to healthy and non-AD subjects) and 327 positive samples that will be used for classification. The discarded scans were kept aside and, together with the samples from the IXI dataset, will be used for autoencoder training. We ensured that the two datasets (for classification and autoencoder training) were disjointed; in other words, the same subject was absent in both datasets. A summary is shown in Table I.

TABLE I
SUMMARY OF THE COMPOSITION OF THE DATASET

	AD	Healthy	Other diseases	Unlabeled
Subjects	173	84	78	692
Scans	327	336	155	2310

IV. METHODS

Our model was inspired by ResNet-18 [31], in which we replaced two-dimensional convolutions with three-dimensional ones and replaced the ReLU activation function with LeakyReLU with a negative slope equal to 0.2. In more detail, the neural network consists of a first convolutional block with 64 filters, a kernel size of 7, stride 2, and a padding of 2. A sequence of four basic block pairs with an increasing number of kernels (64, 128, 256, 512) is then used, each consisting of two pairs of convolution and batch normalization plus a skip connection. All convolutional layers except the first have a kernel size of 3, stride 1, and padding 1. The basic blocks are followed by a global average pooling, a flattening layer, and a dropout layer (with a dropout rate of 0.3), and finally by a classification head consisting of a single sigmoid neuron. A schematic of the model is depicted in Fig. 1.

For unsupervised pre-training, we used the previous model without the classification head as the encoding part of the autoencoder. At the same time, we added a decoder consisting of 5 layers, each with a decreasing number of kernels from 512 to 1. These layers consist of a transposed convolution with kernel size 4 followed by a batch normalization. The first four layers are activated with a ReLU function, and the last with a sigmoid to produce output in the range $[0, 1]$. The stride and padding are calculated so that the size of the output matches that of the input. In addition, a dropout layer with a dropout rate of 0.2 is added as a regularizer in the bottleneck. A schema is shown in Fig. 2.

The advantage of using 3D convolutions lies in their ability to capture spatial information in volumetric data, which is vital in medical imaging applications. 3D convolutions can learn features that reflect relationships between voxels in all three dimensions, unlike 2D convolutions, which can only capture

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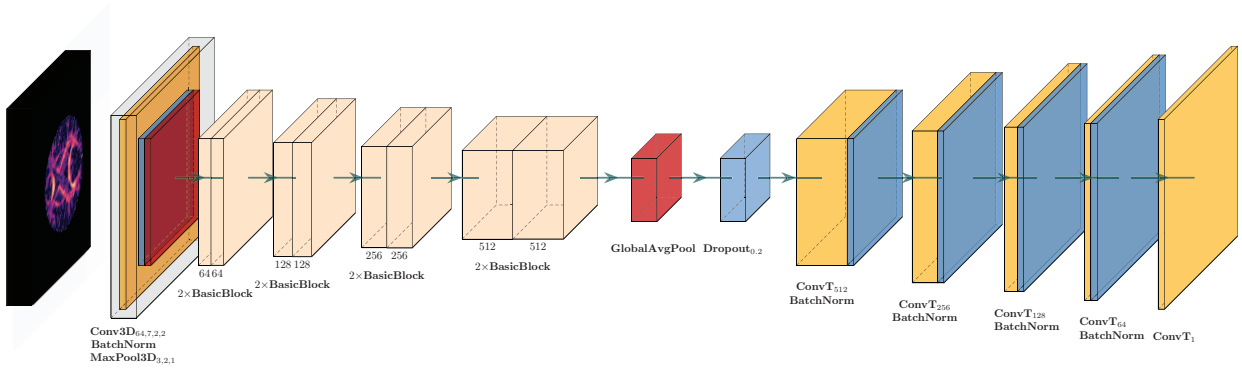

Fig. 2. Schema of the autoencoder model. The meaning of the numbers is the same as in Fig. 1.

spatial information in two dimensions. This allows 3D CNNs to understand data structure better and improve performance in tasks such as AD detection using DTI data.

The rationale behind the use of unsupervised pre-training is that it can help solve the problem of limited labeled data in the medical domain. By pre-training the encoding part of the network on unlabeled data, the model can learn to extract meaningful features from the data without requiring direct supervision. Later, the model can better generalize to new and unseen data by decoupling the decoder and continuing to train only the encoder on labeled data, improving performance. Indeed, unsupervised pre-training can help improve the performance of the subsequent supervised learning task by providing better initialization of network weights.

To overcome the problem of the limited number of samples, we also used online data augmentation techniques, precisely a combination of z-axis mirroring (resulting in hemisphere swapping), three-dimensional rotation, and clipping. These three transformations were applied randomly with a specific probability during training: mirroring was applied with a probability of 0.5, rotation with 0.75, and clipping with 0.8.

Finally, further preprocessing was required to extract fractional anisotropy from the DWI before training the models. First, the scans were scaled to match a common resolution of $2.50\,\mathrm{mm/px}$ in the three dimensions and rotated according to the RPS direction (left to right, anterior to posterior, inferior to superior). Moreover, a brain mask was extracted using the Otzu thresholding method followed by a dilation operation; this mask was used to separate the brain volume from the background. Then, the ordinary least squares method was used with the corresponding *b*-values and *b*-vectors to fit the diffusion tensor (a 3×3 symmetric matrix) to each voxel in order to obtain the diffusion tensor image. From the DTI obtained so far, the fractional anisotropy was calculated as:

$$FA = \sqrt{\frac{1}{2}\frac{(\lambda_1 - \lambda_2)^2 + (\lambda_1 - \lambda_3)^2 + (\lambda_2 - \lambda_3)^2}{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}}$$

where λ_i are the eigenvalues of the diffusion tensor. As a final step, the images were cropped to a size of $96 \times 96 \times 64\mathrm{px}$. This measurement and the chosen resolution represented a volume of $1470\,\mathrm{cm}^3$ sufficient for the average brain volume of $1270\,\mathrm{cm}^3$ and $1130\,\mathrm{cm}^3$ for men and women respectively. Since the values of the FA voxels are within the range $[0, 1]$, scaling was not necessary.

V. Experiments

The experiments were run on Kaggle. The model architectures were implemented using the PyTorch and PyTorch Lightning libraries, while data augmentation was achieved with the Kornia [32] and TorchIO [33] libraries. Instead, all preprocessing steps were performed using Dipy [34].

The classification model was evaluated using a *stratified* 10-fold cross-validation, taking care that the same subject was not present in both training and validation sets, as this could have led to overly optimistic results. The evaluation procedure consisted of two steps: training the autoencoder on the unlabeled samples and performing cross-validation on the classifier by fine-tuning only the encoder. As a performance measure, we used the following metrics:

- Accuracy: the fraction of correct classifications among all classifications;
- Specificity: the fraction of samples correctly identified as negative;
- Sensitivity: the fraction of samples correctly identified as positive;
- AUC: used as a summary of the ROC curve.

For the autoencoder, we used the L1 loss function for 100 epochs with an initial learning rate of the Adam optimizer of 5×10^{-4} and the OneCycle learning rate scheduler [35]. We also experimented with a *denoising* version of the autoencoder, trained to take a noisy version of the input data and aimed at reconstructing the clean input. Specifically, the images were corrupted by Gaussian noise sampled from a normal distribution with mean 0 and random variance between 0 and 0.25. For classification, we adopted the binary cross-entropy as the loss function; in particular, to improve numerical stability, we opted for the loss variant that already includes the sigmoid layer and can exploit the log-sum-exp trick. The loss function was minimized by using the AdamW optimizer with a learning

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TABLE II
RESULTS OF 10-FOLD CROSS-VALIDATION (STANDARD DEVIATION IN PARENTHESES)

	Without pre-training	Pre-training (autoencoder)	Pre-training (denoising autoencoder)
Accuracy	0.634 (0.09)	0.667 (0.05)	0.660 (0.06)
Sensitivity	0.675 (0.20)	0.719 (0.19)	0.683 (0.23)
Specificity	0.604 (0.25)	0.627 (0.18)	0.643 (0.22)
AUC	0.739 (0.08)	0.749 (0.05)	0.750 (0.06)

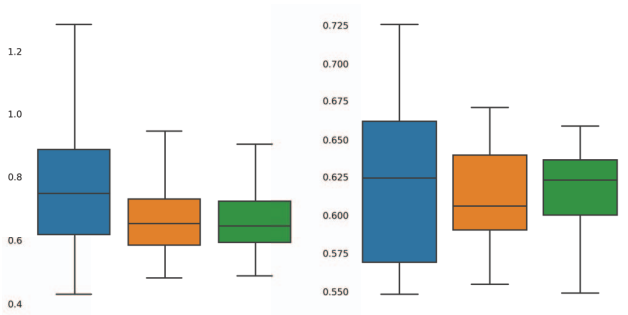

Fig. 3. Box plots of binary cross-entropy loss values measured during training (left) and test (right). In blue, the losses obtained with the model without pre-training; in orange, the model with pre-training; in green, the model with pre-training using the denoising version of the autoencoder.

rate of 5×10^{-5}(5×10^{-3} in the case of fine-tuning) and a weight decay of 0.05. The training process with fine-tuning was developed as follows: the feature extractor was frozen entirely for the first 17 epochs, only the last layer (the last basic block pair) was unlocked, the learning rate was reduced by a factor of 0.1, and training was continued; note that the batch normalization layers were kept frozen throughout training. The model was trained for a maximum of 50 epochs. An early stopping criterion was used only when there was no transfer learning; this criterion monitored the loss computed on a validation set extracted by sampling 0.1% from the training set. A patience of 27 epochs was set.

Table II shows the results obtained. It can be seen that the use of unsupervised pre-training outperforms the baseline method (without pre-training) in terms of overall metrics. This is also reflected in the loss behavior, which is much more stable after unsupervised pre-training (see Fig. 3). The 3D autoencoder produced more stable results (with a lower standard deviation during cross-validation) than the denoising autoencoder. If we take into account that sensitivity is of primary importance in the specific domain, the traditional autoencoder is the suggested strategy for pre-training a feature extractor.

Although these results are promising, it is essential to note that they do not match the state-of-the-art performance obtained with other methods or modalities (generally, higher than 0.85 for all metrics). However, this work represents a first step toward developing better models for AD diagnosis using diffusion imaging and deep learning. It is worth noting that using DTI data in a deep learning context presents many challenges, including the complex nature of the data, noise, artifact issues, and the need for large amounts of annotated data. As we have done in this work, the need for annotation can be alleviated using unsupervised pre-training. Finally, it should be noted that, as mentioned earlier, the positive class in this study consisted of patients with even mild or moderate signs of decline, which may make classification more difficult. In general, the earlier the stage of AD, the more difficult it is to detect it with diagnostic methods.

VI. CONCLUSION

This paper proposed a new approach to Alzheimer's disease detection using diffusion tensor imaging and deep learning techniques. The approach combines unsupervised and supervised learning methods, using a 3D convolutional autoencoder for unsupervised pre-training and a 3D convolutional neural network for supervised learning. The method was evaluated on the OASIS-3 dataset, yielding promising results.

Future work could include evaluating the proposed method on more extensive and diverse datasets to validate its effectiveness further. In addition, interpretation of the learned representations could provide insights into the mechanisms underlying AD and contribute to developing more targeted and personalized treatment options. The method could also be extended to other diseases, such as Parkinson's or multiple sclerosis, to improve early diagnosis. Other DTI modalities, such as axial and radial diffusivity, may provide additional information about the integrity of white matter in the brain. The proposed approach could be extended to provide a more comprehensive analysis of DTI data. Finally, using unsupervised pre-training may be advantageous to alleviate the need for labels that may be difficult to collect in the medical field. This approach could be further explored to improve the robustness and generalization of the proposed approach, especially when dealing with small and unbalanced datasets.

It is worth noting that one of the main limitations of machine/deep learning-based methods to support the detection of AD, related to the relative non-specificity of MRI and DTI for this neurodegenerative disease, is the lack of a complete set of information to ensure diagnosis (in fact, for AD, post-mortem histological examination remains the only way to be sure of the diagnosis). Therefore, research is ongoing to develop frameworks based on more complete information sets.

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